<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>9</td><td>(a)</td><td></td><td></td><td>Indicative content
• lithium fizzes and moves around the surface of the water
• sodium moves faster on the surface, fizzes more and melts into a ball
• potassium reacts more vigorously again, melts into a ball and ignites producing a lilac flame
• reactions more vigorous on moving down the group

• outer electron is lost during the reaction
• lost more easily on moving down the group because it is further away from the nucleus / attraction between the nucleus and the outer electron decreases</td><td>6</td><td></td><td></td><td>6</td><td></td><td>3</td></tr>
<tr><td></td><td></td><td></td><td></td><td colspan="7">5-6 marks
Detailed description of reactions; explanation of relative ease of loss of outer electron
There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.

3-4 marks
Basic description of reactions; reference to loss of outer electron
There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.

1-2 marks
Basic description of some reactions
There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.

0 marks
No attempt made or no response worthy of credit.</td></tr>
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<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>18.8 g (3)

if answer incorrect credit each correct step in one of two possible methods (ecf possible throughout)

method 1

$n(K) = \frac{15.6}{39} = 0.4$ mol (1)

$n(K_2O) = \frac{0.4}{2} = 0.2$ mol (1)

mass $K_2O = 0.2 \times 94 = 18.8$ g (1)

method 2

$M_r(K_2O) = 94$ / mass of 156 (for K)(1)

(156 g K produces) 188 g K_2O (1)

15.6 g K produces 18.8 g K_2O (1)</td><td></td><td>3</td><td></td><td>3</td><td>3</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>3.0×10^{22} (2) accept 3×10^{22}

if answer incorrect award 1 mark for 0.30×10^{23}</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 9 total</td><td>6</td><td>5</td><td>0</td><td>11</td><td>5</td><td>3</td></tr>
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